Fiscal Year:	FY 2005	**Task Last Updated:**	FY 11/01/2005
PI Name:	Brainard, George C. Ph.D.		
Project Title:	Optimizing Light Spectrum for Long Duration Space Flight		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Human Performance Factors, Sleep, and Chronobiology Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	george.brainard@jefferson.edu	**Fax:**	FY 215-923-7588
PI Organization Type:	UNIVERSITY	**Phone:**	215-955-7644
Organization Name:	Thomas Jefferson University		
PI Address 1:	Light Research Program		
PI Address 2:	1025 Walnut St., Room 507		
PI Web Page:			
City:	Philadelphia	**State:**	PA
Zip Code:	19107-5083	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	09/01/2004	**End Date:**	08/31/2008
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	2
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Glickman, Gena (Thomas Jefferson University) Smith, Peter (University of Arizona) Hanifin, John (Thomas Jefferson University) Pineda, Carissa (Thomas Jefferson University) Lockley, Steven (Brigham and Women's Hospital)		
Grant/Contract No.:	NCC 9-58-HPF00403		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The main goal of this project is to determine the optimal wavelengths of light for use as a countermeasure to sleep and circadian disruption for astronauts. Sleep deficits in astronauts are considered a risk factor during space missions. The resulting physiological and behavioral changes can threaten the success of a mission by diminishing alertness, cognitive ability, and psychomotor performance. Over 45% of all medications taken in space are sleep aids. Ambient light is the primary stimulus for circadian regulation in humans. While on Earth humans have a 24-hour day/night cycle to maintain healthy circadian entrainment. In space, astronauts must contend with either a rapidly changing or severely disrupted day/night cycle, and must work in a spacecraft interior that is usually dimly lit and with few or no windows. Bright white light has already been implemented as a pre-launch countermeasure on the ground, but has yet to be used during space flight. Providing sufficient light intensities to work areas in the space shuttle and on the International Space Station (ISS) raises several concerns, among which are heat production and energy consumption. One potential way to mitigate these factors requires first a better understanding of how photic input regulates the human circadian system, and to characterize the effectiveness of this input for different wavelengths and wavelength combinations of light. From this data it may be possible to optimize the light spectrum both as a countermeasure for sleep and circadian disturbances before or during space flight. This four-year project has three specific aims: 1) Test the hypothesis that the white polychromatic fluorescent lamps currently used by NASA as a pre-launch countermeasure for circadian and sleep disruption will have increased efficacy for melatonin suppression when they are enriched in the blue portion of the spectrum. Our published action spectrum indicates that the blue monochromatic wavelengths (446-477 nm) are the most potent for melatonin suppression. This aim will determine if increasing the power in the blue region of a broad spectrum, polychromatic fluorescent light can increase potency for melatonin suppression while keeping the visual experience white. Working collaboratively with Philips Lighting, B.V., we have designed and implemented an operational polychromatic lighting exposure system. Over 50 melatonin suppression trials have been completed with healthy male and female subjects. 2) Test the hypothesis that the white fluorescent lamps currently used by NASA as a pre-launch countermeasure for circadian and sleep disruption will have increased efficacy for circadian phase-shifting when they are enriched in the blue portion of the spectrum. To achieve this aim, we plan on testing the magnitude of the phase-shift of melatonin and core body temperature circadian rhythms in healthy subjects following exposure to equal photon densities of blue-enriched polychromatic white fluorescent lights against the current fluorescent lights currently used by NASA as a pre-launch countermeasure. The results of this study will help specify the onboard lighting system to be used on future long duration, exploration-class missions. This aim is dependent on the results from Aim 1, and will be initiated during the coming research year. 3) Test the hypothesis that three ambient Martian light environments (clear midday skylight, clear sunrise skylight, and overcast sunrise skylight) will differ in their capacity to regulate melatonin secretion. Since circadian and sleep regulation depends on an astronaut's total exposure to both artificial as well as passive ambient illumination through windows or space suit visors, it is essential to determine the circadian effects of light exposures in non-terrestrial environments. To test this hypothesis, we are working collaboratively with the Lunar and Planetary Laboratory investigators who extensively characterized the daylight spectrum and intensity at the surface of Mars during the Pathfinder Mission in 1997. Together, we are developing a polychromatic exposure system that simulates the spectral characteristics of Martian daylight. We will use this apparatus to establish three melatonin suppression fluence-response curves with polychromatic light that simulates Martian light environments. We have traveled to the Lunar and Planetary Laboratory in Tucson, Arizona and have ordered the necessary equipment to initiate this aim. There has been a manufacturing delay which has caused the Lunar and Planetary Laboratory to request additional time to fabricate this equipment without additional funding. In the coming year, we plan on completing the second of three fluence-response curves for melatonin suppression with blue enriched prototype lamps, as well as having a substantive start on the third fluence-response curve. If the trials succeed, these lamps may provide NASA with their first deliverable lighting countermeasure to sleep and circadian disruption for use during space flight. We also plan to initiate subject recruitment to determine the circadian phase-shifting ability of prototype polychromatic "blue enriched" fluorescent lamps. Finally, we plan the completion of the first Martian light exposure system. Data from these studies will impact NASA research objectives and technology requirements to improve light treatment as a countermeasure for circadian and sleep-wake disruption in NASA space flight missions, identify the best spectral transmission for space suit visors and the windows used in space vehicles and habitats, and engineer the ideal spectral distribution for illumination of general living quarters during space exploration. If the trials using blue enriched polychromatic lamps succeed, these lamps may provide NASA with their first deliverable lighting countermeasure to sleep and circadian disruption for use during space flight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	A significant portion of the global population suffers from chronic sleep loss and/or circadian-related disorders. Evidence for disease or illness to occur due to a disruption of circadian homeostasis has mounted significantly in the past several years. In the United States alone, some 20 million Americans work shifts which do not allow them a biologically healthy nocturnal sleep cycle. This group has been shown to be more likely to suffer from a wide variety of ailments, including cardiovascular disease, gastrointestinal distress, cognitive and emotional problems. Furthermore, a recent epidemiological study of female night-shift nurses has shown that they are statistically more likely to suffer from breast cancer and colon cancer compared to day shift workers. Aside from evidence for a breakdown in physical health, the effects of circadian disruption and sleep loss have long been known to have potentially dangerous behavioral effects. Mental fatigue, diminishment in alertness, loss of psychomotor coordination and physical performance are all commonly found in individuals with sleep loss, sleep debt, or circadian misalignment. The impact of these dangers affects many industries, including transportation, manufacturing, communications, and medicine. It has long been a source of concern for the military, as well. Additionally, many people experience the same effects due to jetlag from air travel across several time zones. Understanding the biological impact of ambient light on the human circadian system is a critical step toward the eventual application of countermeasures to these and other sleep and circadian-related disorders in civilian populations. Quantifying the relative potency of different wavelengths of light for melatonin suppression via the retinohypothalamic tract will provide key insights not only into the basic underlying physiology of this photosensory system, but also produce guidelines for how a misalignment of the system could be most effectively corrected. This research is also expected to have a wide-ranging impact on future types of artificial indoor lighting and architectural illumination in general. The lighting industry has taken notice of the biological impact different types of light is capable of having, and has already begun to invest in furthering our understanding of light's role for human health. There are a number of existing therapeutic interventions using light that stand to benefit from an enhanced understanding of how specifically different wavelengths of the spectrum affect the human circadian system. A more efficient intervention, with increased potency and/or less side-effects could result. One such disorder currently being treated with bright white light is Seasonal Affective Disorder (SAD), also known as winter depression. It is estimated that as many as 1 in 5 Americans suffer from SAD or its milder version, subsyndromal Seasonal Affective Disorder (sSAD). Side effects from exposure to

	bright white light for this and other therapies include: hypomania, headache, vision problems, nausea, dizziness, and anxiety. Similar bright white light interventions are also used for treating jetlag. Optimizing the light spectrum for specific affective and/or circadian-related disorders could deliver the same biological impact with lower levels of light intensity, and potentially less side-effects.
Task Progress:	In collaboration with the HPFSC team, Philips Lighting, B.V. has continued to design and provide prototype lamps in order for us to conduct experiments and build fluence-response curves for melatonin suppression using “blue-enriched” polychromatic light. “Blue-enriched” polychromatic light will theoretically provide a stronger circadian stimulation but also support vision. A total of over 50 subject/study nights related specifically to this NSBRI funded project have been completed within the past research year. This required the recruitment and screening of 17 subjects. To be eligible potential subjects must have consistent sleep/wake cycles, normal color vision, good ocular and psychological health, and be drug free. These parameters are met through wearing an actigraph, a watch-size device measuring movement, taking the Farnsworth-Munsell 100 Hue Test, undergoing an ocular and psychological examination, and having a urine drug analysis. After screening, 10 subjects entered the study. Each subject was scheduled for a series of ten exposures with at least one week of rest between study nights. Of the 50 subject/study nights, all samples have been assayed for melatonin content by radio-immunoassay. The completion of data gathering is currently underway. Also, within the past research year, we have implemented a four hour adaptation period of dim light into the design of our study. This new step required altering our approved human use (IRB) protocols. Thomas Jefferson University’s IRB committee ultimately reviewed and approved this new protocol. The adaptation period was initiated due to the evolution of the circadian field based on both published and unpublished data yielding more accurate melatonin suppression results. Relevant to Specific Aim 1, two studies have began initiation within the past research year. Ten subjects are in the process of completing a melatonin suppression study using polychromatic light. Eight of those subjects are more then half-way through a series of ten exposures to polychromatic blue-enriched fluorescent light. The remaining two subjects have begun a similar study using a polychromatic light source that is not “blue-enriched”. In order for Specific Aim 2 to be initiated, results from Specific Aim 1 must first be obtained. In the meantime, we have solidified our collaboration with Dr. Steven Lockley of Harvard University for our work on the phase-shifting capability of blue-enriched polychromatic lights. Experimental work on Specific Aim 3 has been initiated this research year but is behind schedule due to a delay in equipment fabrication on the part of the Lunar and Planetary Laboratory. Consequently, the Lunar and Planetary Laboratory has requested additional time without additional funding to complete their component of this project. Barring further unforeseen delays, we plan on building and initiating testing on the Mars lighting system.
Bibliography Type:	Description: (Last Updated: 01/19/2026)
Articles in Peer-reviewed Journals	Brainard GC, Hanifin JP. "Photons, clocks, and consciousness." J Biol Rhythms. 2005 Aug;20(4):314-25. Review. PMID: 16077151 , Aug-2005
Articles in Peer-reviewed Journals	Fucci RL, Gardner J, Hanifin JP, Jasser S, Byrne B, Gerner E, Rollag M, Brainard GC. "Toward optimizing lighting as a countermeasure to sleep and circadian disruption in space flight." Acta Astronaut. 2005 May-Jun;56(9-12):1017-24. PMID: 15838948 , May-2005
Awards	N/A "Jefferson Medical College Class Day Speaker (Voted favorite professor by graduating medical students), 2005" Jan-2006
Awards	N/A "Jefferson Medical College Dean's Award for Distinguished Teaching in Basic Science, 2005" Jan-2004
Books/Book Chapters	Brainard GC, Hanifin JP. "The effects of light on human health and behavior: relevance to architectural lighting." in "Light and Health: non-visual effects ; Proceedings of the CIE Symposium 04, 30 Sep. - 2 Oct. 2004, Vienna, Austria." Vienna: CIE (Commission Internationale De L'eclairage), p. 2-16, 2004. CIE publication x027:2004., Dec-2004
Presentation	Brainard, G. C. "Circadian Phototransduction: Basic Sciences to Clinical Trials" N/A Apr-2005
Presentation	Brainard, G. C. "Circadian Phototransduction: Significance to Light Therapy and Manned Space Flight" N/A Feb-2005
Presentation	Brainard, G. C. "Combating Sleep Loss and Shift Work Problems" N/A Mar-2005
Presentation	Brainard, G. C. "Optimizing Light for Human Space Exploration" N/A May-2005
Presentation	Brainard, G. C. "Optimizing Light for Long Duration Space Flight" N/A Feb-2005
Presentation	Brainard, G. C. "Optimizing Light Spectra for Long Duration Space Flight" N/A Jan-2005
Presentation	Brainard, G. C. "The Biological Potency of Light in Humans: Significance to Health and Behavior" N/A Sep-2004
Presentation	Brainard, G. C. "The Biological Potency of Light in Humans: Significance to Health and Behavior" N/A Oct-2004
Presentation	Brainard, G. C. "The Impact of Light on Health" N/A Jul-2004